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ORIGINAL ARTICLE

# Effects of Alcoholic Beverage Control Policies and Contextual Factors on Alcohol Consumption and its Related Harms in France From 1960 to 2000

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Important social changes such as urbanization, increases in female education and employment, and increased incomes have occurred in France from 1960 to 2000 along with a major decrease in alcoholic beverage consumption (from 25 L pure alcohol per inhabitant 15 years and old to 13 L); especially due to wine consumption decrease. These changes in drinking patterns are associated with significant decreases in alcohol consumption-related harms (liver disease mortality and transport accident mortality). Several alcoholic beverage consumption control policy measures were also created during this period. This study explores the impact of these policies measure on alcohol consumption and alcohol consumption-related harms, adjusted with selected social changes. France's control policy has been associated, partially, with regressive effects on alcohol consumption but not on alcohol consumption-related harms. Study limitations are noted.

**Keywords** historical background and changes in consumption, drinking patterns and harm from 1960 to 2000

## HISTORICAL CONTEXT

Between 1960 and 2013, France has experienced two distinct economic phases. First, the end of the postwar economic boom (1945–1973), which was an uninterrupted period of economic prosperity. The French entered the consumer society, associated with increased building in public and private housing, training and education. Second, since the 1973 oil crisis and the economic crisis that resulted, the economy entered a period of slow growth, a

phenomenon that has increased with globalization and the more recent economic recession. If the average standard of living continued to rise, a recurrent mass unemployment and a widening of social inequalities reappeared. At a sociological level, changes of the French population are considerable. The still very rural population of the 1960's has increased from 45 to 65 million people and is now 75% urban. Several major developments are noteworthy: the dramatic increase in life expectancy and 82.8 years old, respectively, in 2000, the development of women's employment, the sharp decline in primary and secondary sectors in favor of tertiary sectors (in 1900, the tertiary sector represented 30% of employments in France while it was 70% of the active population one century later: *France, a previously highly agricultural and industrial country, has evolved into an economy of service*), and finally a significant increase in the average level of education.

Since the early 1960s, France has seen a gradual and sustained decrease in alcoholic beverage consumption per capita, per year in persons over age 15 years. This decrease, related mainly to a diminished wine intake, is to the credit of public health policies<sup>1</sup> which have changed the image of alcohol, which is now perceived as being a product that may cause health risks. This decrease in alcohol consumption is, however, accompanied by changes in drinking patterns as shown by surveys conducted during the past 20 years. There has been an increase in consumption by young people, a less significant difference among genders and episodic or chronic excessive drinking among men over 40 years, which remain issues of concern.

Participant organizations in AMPHORA can be seen at <http://www.amphoraproject.net>.

<sup>1</sup>The reader is referred to Hill's criteria for causation which were developed in order to help assist researchers and clinicians determine if *risk factors* were causes of a particular disease or outcomes or merely associated. (Hill, A. B. (1965). The environment and disease: associations or causation? *Proceedings of the Royal Society of Medicine* 58: 295–300.). Editor's note.

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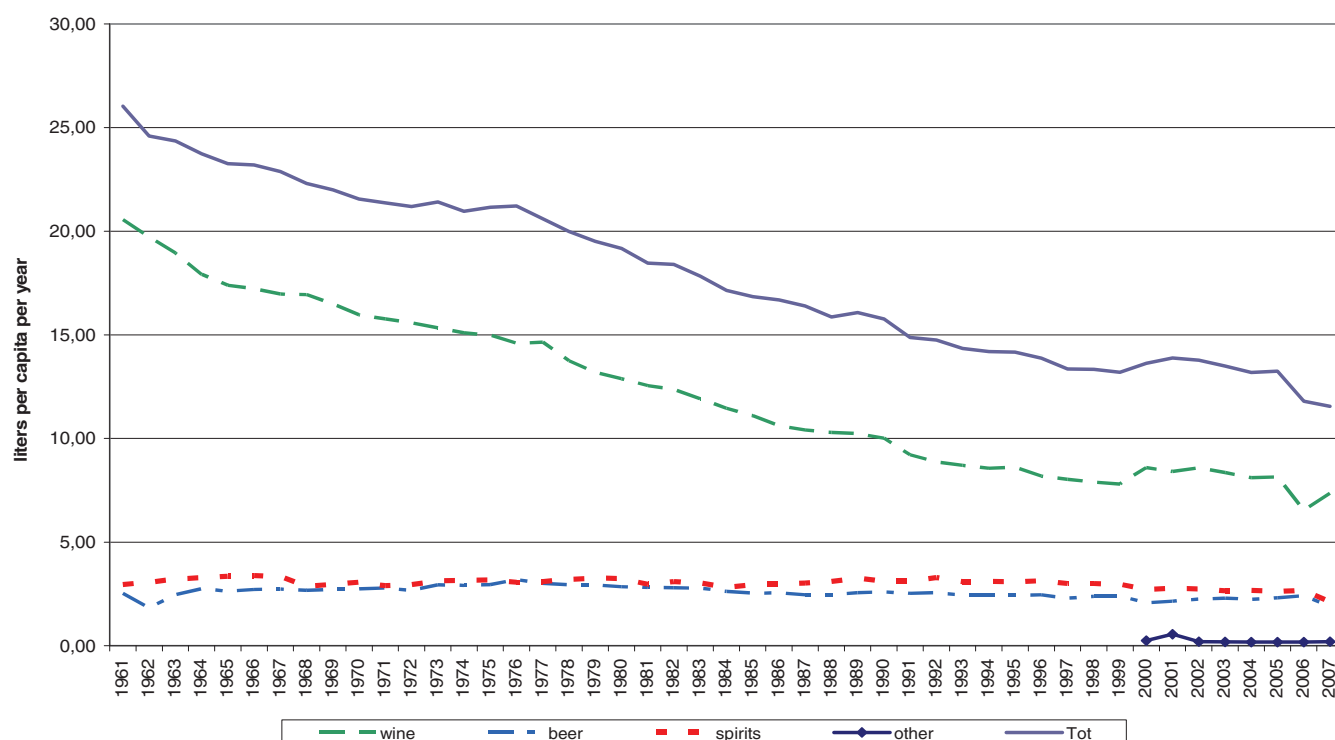


FIGURE 1. Alcohol recorded consumption in France, liters of pure alcohol per inhabitant 15 years and older, 1961–2007—source: WHO GISAH, 2012\*. \*Data were collected through AMPHORA work package 2, thanks to Jurgen Rehm and Margaret Rylett. Other = all wines that are not grape wine, all beers not made from malt, cocktails, mixed drinks, fortified wines, cider, and the like. NB = for some years the sum of total alcohol consumption, and the sum of the different types of beverages may not correspond.

Multiple measures expressed in the Labor, Traffic, Tourism, Tax, and Public Health Codes have been implemented. Among the changes in the law are the prohibition on driving with a BAC greater than or equal to .8 g/L in 1970, with a further reduction to .5 g/L in 1995. In addition new media controls include: (1) the prohibition advertising alcoholic beverages on television, (2) the prohibition of advertising of alcoholic beverages in publications aimed at young people, and (3) the obligation to accompany any advertising with a health message stating that Alcohol abuse is dangerous for health. Moreover, France is the only European country, since 2007, where all alcoholic beverages have a compulsory health warning or pictogram recommending no consumption of alcohol during pregnancy.

Despite the decrease in alcoholic beverage consumption in France since the 1960s, the decline in mortality from liver disease, and the decrease in road accidents which can be partially associated with measures designed to effect drinking and driving, alcoholic beverage consumption remains a contributor of morbidity and premature mortality. Alcoholic beverages are France's second leading cause of preventable death after tobacco smoking (Figure 1).

## CONSUMPTION

Data available from 1961 to 2007 on per capita recorded alcoholic beverage consumption in those aged 15 years and older are provided by GISAH (Global Information Sys-

tem on Alcohol and Health) (World Health Organization (WHO), 2012), through AMPHORA Work package 2.

As reported in the Table 1 and in the Figure 1, total alcohol consumption, and the sum of the different types of beverages, which was subsequently calculated, may differ in France by .02–.37 L of pure alcohol during most years. This also holds true for 2009 GISAH dataset, which show the same differences for the period 1961–2005. Possibly, these differences may be related to the process of converting from the per capita of total population to the per capita of 15 years and over. However, we considered that this was a minor problem since in most of the analysis the GISAH figures of per capita alcoholic beverage consumption were used.

During the 1950s, France had its postwar peak in alcoholic beverage consumption (the amount of pure alcohol consumed for the *whole population* was 19.81 in 1951, compared with 18.58 in 1965 and 16.96 in 1975, World Drink Trends, 2002). A decrease began since then, which is also observable at the start of our study period. In 1961, total alcoholic beverage consumption was 26.03 L of pure alcohol per capita of persons 15 years old or more. A constant decreasing trend (substantially due to wine) reached 11.55 L in 2007; a 56% decrease.

France is traditionally a “wine country.” Wine consumption represented 79% of the total alcoholic beverage consumption in 1961 (when 20.56 L was the pure alcohol per capita intake). It decreased to 64% in 2007 (7.35 L). During the same period, beer consumption also decreased by a slower steady pace of 26%, while spirits consumption

TABLE 1. Alcohol recorded consumption in France, liters of pure alcohol per inhabitant 15 years and older, 1961–2007—source: WHO GISAH, 2012\*

| Country | Year | Wine  | Beer | Spirits | Other | Tot   |
|---------|------|-------|------|---------|-------|-------|
| France  | 1961 | 20.56 | 2.53 | 2.95    |       | 26.03 |
|         | 1962 | 19.71 | 1.82 | 3.07    |       | 24.59 |
|         | 1963 | 18.94 | 2.47 | 3.22    |       | 24.34 |
|         | 1964 | 17.91 | 2.76 | 3.29    |       | 23.73 |
|         | 1965 | 17.39 | 2.63 | 3.35    |       | 23.25 |
|         | 1966 | 17.23 | 2.71 | 3.38    |       | 23.20 |
|         | 1967 | 16.97 | 2.73 | 3.35    |       | 22.88 |
|         | 1968 | 16.94 | 2.67 | 2.87    |       | 22.30 |
|         | 1969 | 16.48 | 2.71 | 2.96    |       | 21.99 |
|         | 1970 | 15.96 | 2.75 | 3.06    |       | 21.55 |
|         | 1971 | 15.77 | 2.78 | 2.89    |       | 21.37 |
|         | 1972 | 15.58 | 2.67 | 2.95    |       | 21.19 |
|         | 1973 | 15.34 | 2.94 | 3.12    |       | 21.41 |
|         | 1974 | 15.09 | 2.91 | 3.15    |       | 20.96 |
|         | 1975 | 14.99 | 2.95 | 3.18    |       | 21.16 |
|         | 1976 | 14.59 | 3.19 | 3.06    |       | 21.21 |
|         | 1977 | 14.64 | 3.01 | 3.09    |       | 20.60 |
|         | 1978 | 13.75 | 2.94 | 3.19    |       | 19.99 |
|         | 1979 | 13.19 | 2.94 | 3.27    |       | 19.51 |
|         | 1980 | 12.88 | 2.85 | 3.24    |       | 19.17 |
|         | 1981 | 12.55 | 2.82 | 2.96    |       | 18.46 |
|         | 1982 | 12.37 | 2.79 | 3.09    |       | 18.40 |
|         | 1983 | 11.91 | 2.78 | 3.03    |       | 17.83 |
|         | 1984 | 11.45 | 2.62 | 2.82    |       | 17.14 |
|         | 1985 | 11.10 | 2.54 | 2.95    |       | 16.84 |
|         | 1986 | 10.62 | 2.55 | 2.96    |       | 16.68 |
|         | 1987 | 10.42 | 2.45 | 3.03    |       | 16.39 |
|         | 1988 | 10.28 | 2.47 | 3.11    |       | 15.86 |
|         | 1989 | 10.24 | 2.56 | 3.25    |       | 16.08 |
|         | 1990 | 10.01 | 2.60 | 3.12    |       | 15.77 |
|         | 1991 | 9.21  | 2.53 | 3.11    |       | 14.88 |
|         | 1992 | 8.87  | 2.56 | 3.29    |       | 14.75 |
|         | 1993 | 8.71  | 2.44 | 3.09    |       | 14.34 |
|         | 1994 | 8.56  | 2.45 | 3.10    |       | 14.20 |
|         | 1995 | 8.61  | 2.43 | 3.08    |       | 14.16 |
|         | 1996 | 8.18  | 2.45 | 3.14    |       | 13.88 |
|         | 1997 | 8.02  | 2.29 | 3.00    |       | 13.35 |
|         | 1998 | 7.89  | 2.38 | 3.00    |       | 13.33 |
|         | 1999 | 7.80  | 2.39 | 2.96    |       | 13.19 |
|         | 2000 | 8.59  | 2.08 | 2.71    | 0.25  | 13.63 |
|         | 2001 | 8.40  | 2.15 | 2.77    | 0.56  | 13.89 |
|         | 2002 | 8.59  | 2.25 | 2.74    | 0.20  | 13.78 |
|         | 2003 | 8.36  | 2.29 | 2.65    | 0.19  | 13.49 |
|         | 2004 | 8.10  | 2.24 | 2.67    | 0.17  | 13.18 |
|         | 2005 | 8.14  | 2.31 | 2.62    | 0.17  | 13.24 |
|         | 2006 | 6.54  | 2.42 | 2.67    | 0.17  | 11.80 |
|         | 2007 | 7.35  | 1.88 | 2.13    | 0.20  | 11.55 |

\*Data were collected through AMPHORA work package 2, thanks to Jurgen Rehm and Margaret Rylett.

Other = all wines that are not grape wine, all beers not made from malt, cocktails, mixed drinks, fortified wines, cider, and the like.

NB = for some years the sum of total alcohol consumption, and the sum of the different types of beverages may not correspond.

manifested an up and down trend, ending with a smaller drop of 9% (from 2.95 L of pure alcohol per capita in 1961 to 2.13 in 2007); spirits' consumption placing second after wine.

## Drinking Pattern and the Public Concern About Alcohol

The decrease in alcoholic beverage consumption since 1960 is associated with a change in France's consumption patterns. According to professionals, alcoholic beverage consumption evolves from drinking it everyday, at every meal, as a "food" to a more festive status. These findings are confirmed by data from INSEE: the French are turning to drinking quality wines (2.7% increase per year on average since 1960) at the expense of ordinary wine consumption (2.4% decline per year) (Legleye & Beck, 2007).

## Detrimental Consequences

*Liver disease and liver cirrhosis* (or simply liver disease) mortality indicates the extent of high and continuous alcohol consumption, whereas the number of persons killed or harmed in *transport accidents*, associated with alcohol consumption, indicates single intoxication events (World Health Organization, 2012; Rehm & Scafato, 2011). The figures of both types of detrimental consequences of alcoholic beverage consumption were standardized by age and by gender by Sarah Forberger (AMPHORA work package 2) and are presented below. The series for both mortality rates are for the period 1980–2009.

However, WHO Health For All Database (HFA DB) provides mortality standardized death rates (SDR) for both genders and for all ages per 100,000 inhabitants (France has complete liver mortality series for 1970–2008; and complete transport mortality series for 1979–2008). (World Health Organization, 2010). This material will be used in the analysis since the figures are more comparable with alcoholic beverage consumption data at all ages and for both genders.

Figure 2 shows a remarkable 1/4 drop in mortality due to liver disease and cirrhosis in France, from about 35 deaths per 100,000 male inhabitants (15.6 for females) in 1979 to 8.9 in 2008 (females 3.3).

By and large, the decrease in liver deaths in France is parallel to the decrease in alcohol (wine) consumption, which has occurred between 1960 (and in the previous decade), and the 2000s (see consumption figures in Table 1 and Figure 1). Considering results by age, we clearly see that liver disease mortality has considerably decrease in 40–64 years old males (from 82.9 to 21.5).

Figure 3 shows a drop in mortality due to transport accidents in France by about 1/3 in all ages and both genders, from about 38 deaths per 100,000 male inhabitants (11 deaths for females) in 1979 to 14 (3.2 for females) in 2008. As in the case of liver mortality, the decrease in transport mortality during 1979–2008 is approximately parallel to a similar decreasing trend in consumption (see consumption figures in Table 1 and Figure 1). Differences between males and females are very clear in the figure above as men have much higher mortality rates than women. Moreover, 15–24 years old population is also more exposed to transport accident death.

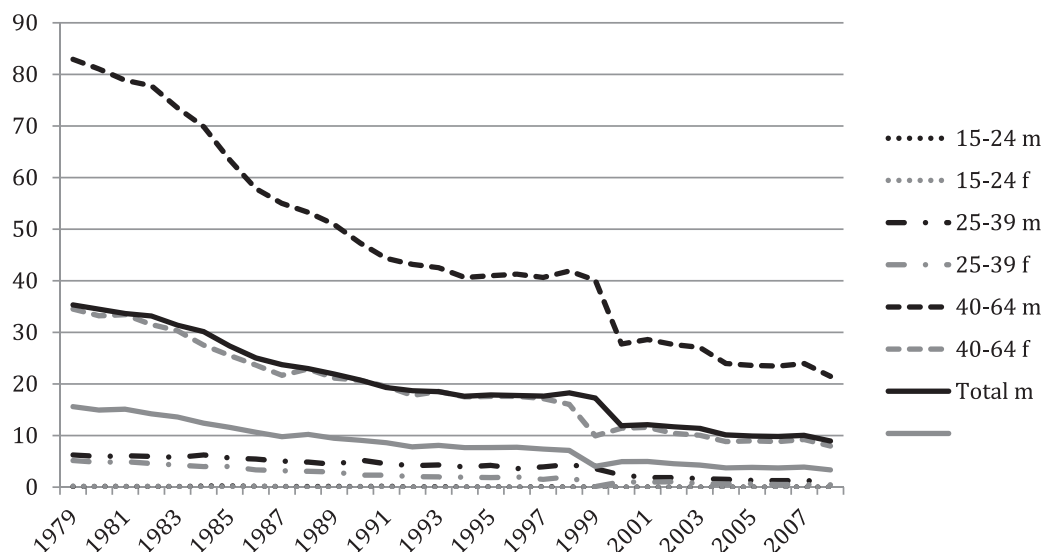


FIGURE 2. Liver disease mortality rate per gender & age (per 100,000 population). France, 1979–2008—Source: WHO mortality data, 2009\*. \*Data were collected through AMPHORA work package 2, thanks to Jurgen Rehm and Sarah Forberger.

## AIM

This study is based on the thesis, empirically supported by the specific literature, that alcoholic beverage consumption control policies alone do not cause nor explain changes in alcoholic beverage consumption and in alcohol consumption-related harm. These changes can only be reasonably explained when other contextual factors, such as social, cultural, economic, religious, demographic, amongst other factors, are considered simultaneously.<sup>2</sup>

This study was designed to identify changes in selected contextual factors that are most strongly correlated with changes in alcohol consumption, drinking patterns, and their related harms between 1960 and 2008 in France.

A country-bound variable combination of the impact of (a) preventive alcoholic beverage control policies and (b) unplanned determinants occurring during the period 1960–2005, that have not been part of alcohol policy measures, would explain the changes in alcoholic beverage consumption and alcohol consumption-related harm, instead of the intervention of alcohol policy alone.

<sup>2</sup>The reader is asked to consider that with the advent of artificial science and its theoretical underpinnings (chaos, complexity, and uncertainty theories), it is now posited that much of human behavior is complex, dynamic, multidimensional, level/phase structured, nonlinear, lawdriven, and *bounded* (culture, time, place, age, gender, ethnicity, etc.). “Control,” however, it is defined and delineated, and with whatever formal as well as informal policies, would be such a process. There are a number of important issues to consider and which are derived from this: (1) Using linear models/tools to study nonlinear processes/phenomena can and does result in misleading conclusions and can therefore also result in inappropriate intervention; (2) the concepts *prediction* and *control* have different meanings and dimensions than they do in the more traditional linear “cause and effect” paradigms; (3) uncertainty, unpredictability and the lack of *real* control, and not just attempts at influencing, are the dimensions of reality. (Buscema, M. (1998), *Artificial Neural Networks*, Substance Use & Misuse, 33(1–3); Ormerod, Paul, (2005) *Why most things fail: Evolution, extinction and economics*. Faber & Faber, UK; Miller, Matt, (2010) *The tyranny of dead ideas*; New York: Henry Holt & Co. Editor’s note.

If the results support this study’s aims, this broader perspective would allow, minimally, to specify which economic, social, and cultural changes operating along with legislated alcohol control policy measures actually influence consumption. However, perhaps of greater importance, a study incorporating such generalizable factors would also allow relevant policy makers to estimate the necessary circumstances under which alcoholic beverage control policies are likely to be more effective, less effective, to be regressive in outcomes, and even to be irrelevant to be planned for needed changes, wasting limited human and nonhuman resources. Such information and useable knowledge could also sensitize policy makers to a range of social, economic, or cultural developments that requires new and more specific prevention initiatives.

## METHODS

Seven unplanned variables with the largest data coverage were considered for the analysis. These included: income, prices of alcoholic beverages, males aged over 65, female educational level, female employment, mother’s age at all childbirths, and urban level.

### Multiple Imputation and Selection of Four Unplanned Variables

Due to the incompleteness of the time series data to be used for the quantitative analysis, a *multiple imputation* of missing values was performed. Even if our main interest was in filling missing entries for the seven variables selected to be included in the statistical model, the multiple imputation procedure involved all of the information collected both from the AMPHORA study partners and the international sources (EUROSTAT, 2011; OECD, 2011; World Bank, 2011a, 2011b, 2011c), concerning both alcoholic beverage consumption and the unplanned factors. Multiple imputation is a useful strategy for dealing

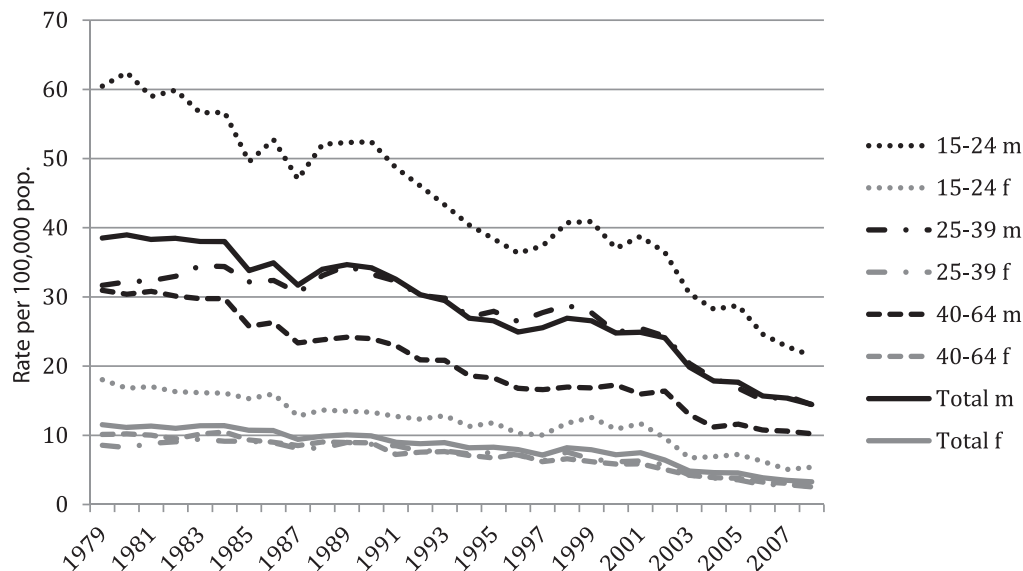


FIGURE 3. Transport accidents mortality rate per gender & age (per 100,000 population). France, 1979–2008. Source: WHO mortality data, 2009\*. \*Data were collected through AMPHORA work package 2, thanks to Jurgen Rehm and Sarah Forberger.

with incomplete data sets by filling in the missing entries with a set of plausible values. Here, a Multivariate Imputation by Chained Equations (MICE) was carried out (Raghuathan et al., 2001).

The imputation required a preliminary step consisting in the selection of a matrix of predictors from all the AMPHORA study partner's national datasets for each variable with missing values. We first filled in the missing data by randomly sampling from the empirical marginal distributions, and then regressed the  $i$ th variable on each of the other chosen variables, selecting as predictors the four variables that generated the smaller AIC (Akaike information criterion). The set of the predictors was enriched with terms which captured the time pattern ( $t$ ,  $t^2$ , and  $t^3$ ). In case of multiple sources of data, if for the  $i$ th variable a time series of the same factor from another database was available, this information was forced to be a predictor. The four variables with the smallest AIC are "Female educational level," "Female employment," "Mothers' age at all childbirths," and "Urban level." These variables were retained for the next analysis. Once the predictors' matrices were defined, the MICE package implemented used R software.

### The Core Model Applied on two Types of Models

The time series structure was explored and we found that an ARIMA first-order integrated model could be appropriate for most countries. However, the results of this model appeared very close to those obtained from a linear regression with a time trend among the explanatory variables. Therefore, we used this second approach, which is simpler and more understandable.

A core model was specified for the logged procapita total alcoholic beverage consumption, accounting for income and alcohol prices (the first two types of beverages most drunk over the study period). This approach rests on the relative importance of economic factors in determin-

ing changes on alcoholic beverage consumption (see e.g. Nelson, 2010). Moreover, the core model also took into account the males over age 65 in the population (percentage of elderly), so that the demographic age structure of the population was considered. All of these variables were included in the linear predictor after a logarithm transformation. A time trend was also added, to capture the long-term consumption behavior that could be related to unobservable factors. An assumption of independent normally distributed error terms was made.

The other four unplanned variables selected for the analysis ("Female educational level," "Female employment," "Mothers' age at all childbirths," and "Urban level") that are socially relevant, were then inserted in the core model one at a time after a log transformation (Nelson, 2010). The AIC for each model was calculated, and the model with a minimum AIC was detected. This was done for the practical reason of avoiding the effects of the high correlation between these variables, which would create very unstable results if they were included in the model at the same time.

Since the effect of these unplanned indicators was expected to be delayed in time, they were lagged, by considering the mean value among the current year's value, and those values in the two preceding years:

$$X_t = \frac{Z_t + Z_{t-1} + Z_{t-2}}{3}. \quad (1)$$

The resulting model is:

$$\log C_t = \beta_0 + \beta_1 t + \beta_2 \log l_t + \beta_3 \log O_t + \beta_4 \log P_t^C + \beta_5 \log A_t + \beta_6 \log X_t + \varepsilon_t. \quad (2)$$

Due to both dependent and explanatory variables being logged, the model's coefficients can be read as elasticity.

The Interval Confidence was set at .90 instead of .95, which is the usual value for CI. This liberal use of CI was used in order to capture even minor information from data.

### ***Association Between Control Policy Measures and Alcohol Consumption***

In order to study the effect of planned indicators, the effect of each policy was initially estimated, one at time, in a model adjusted for the unplanned indicator that provided the minimum AIC, i.e. female employment.

Policy 3 (see *Part C* about Variables collected), which relates to tax variations, was excluded from the analysis, since it might be strongly correlated with alcoholic beverage prices, which have already been included in the model as unplanned indicators.

Since policies 1 and 2 occurred during the same year (1970), a single effect for those policies was estimated.

### ***Association Between Control Policy Measures and Alcoholic Beverage Consumption-Related Harms***

An analysis was made on the effects of the selected control policy measures on harm related to alcohol consumption, which focused on liver mortality and transport mortality. The public health approach considers that restrictive alcohol policies, which are designed to reduce alcohol consumption, should, as a consequence, also be able to produce a reduction in harm. This is the reason why our model considers the *indirect* effect of policy measures on harm, i.e. the effects on harm mediated by consumption

In addition, the *direct* effect of policy measures on harm was also studied. This would allow us to observe the effects on harm of those control measures which may be effective, hypothetically, even when they are not mediated by alcohol consumption. For example, under conditions in which restrictive drinking and driving measure could induce, or be associated with, consumption decrease, or cessation, among drivers who would not drink anymore, or would only drink when not driving, but that would not be capable to cause a global reduction in the consumption of the general population.

In order to study the effect of policy measures (see *Part C* about Policy measures and unplanned variables collected), the effect of each policy was estimated, one at time, in a model adjusted for the unplanned indicator that provided the minimum AIC in France, i.e. *female employment*.

HFA DB, which provides mortality SDR for both genders and all ages per 100,000 inhabitants, was used in this analysis (World Health Organization (WHO), 2009). Liver mortality series are complete for the period 1970–2008, whereas transport accident mortality series are complete for the years 1979–2008. For the data, see *Part A* about historical and drinking context since 1960–2000.

### ***Selected Control Policy Measures and Unplanned Variables***

#### ***Alcoholic Beverage Consumption Control Policies***

A number of control policy measures had been implemented in France before 1960. During the period 1960–2008, about 70 country-wide alcohol consumption policy measures were introduced into the French legislation. A smaller number of policy measures, which was consensually set at 6, were considered appropriate to better analyze the effect of policies within each country and across the 12 participating AMPHORA study countries (see *Section 1, Collection and selection of alcohol policy measure*). In order to achieve this, the study's central team in Florence proposed a list of relevant policy measures which was agreed upon by the French team. The six measures to be used for analysis, all of which are *restrictive* measures, are listed below:

- (1) 1970  $>.8\text{g/L} = \text{drink and driving infraction};$   
 $> 1.2\text{g/L} = \text{offence}$
- (2) 1970 creation of free and anonymous "nutritional hygiene centers" (CHA) on an experimental basis, aiming at early detection and *treatment* for alcohol abusers (without manifesting any additional major psychiatric disorders)
- (3) 1977 *excise* duty increases on beer  $> 4.6^\circ$
- (4) 1987 total *advertising* ban of alcoholic beverages on TV
- (5) 1991 *minimum age*: sale of alcoholic drinks to minors  $< 16$  is forbidden; on-premise sale of alcoholic drinks  $> 15^\circ$  to minors  $> 16$  is forbidden. The alcohol policy law, the Loi Evin, was passed in France during this year, in order to control the advertising of alcohol (and tobacco).
- (6) 1995  $>.5\text{g/L} = \text{drinking and driving infraction};$   
 $> .8\text{g/L} = \text{offence}.$

#### ***Unplanned Variables***

Seven unplanned variables were identified in order to be used for time trend analysis in France, representing social, economic, and demographic determinants of consumption change. They are *Males over age 65*, *Female education and Female employment*, *Mother's average age at all childbirths*, and *Urbanization, Income, and Prices of the two most consumed beverages during the study period in France, i.e. beer and wine*. Figure 4 indicates the rates of each variable over time, on the basis of French data collected from national sources (that refer to *Mothers age at their first childbirth*). To make the data series completed for all of the variables, and to enable the analysis, a multiple imputation was subsequently done on these data also using international sources. The *i*th variable on each of the other chosen variables was then regressed, selecting as predictors the four variables that generated the smaller AIC (see *Part B* about the summary of the aim and the methods used). The four variables with the smallest AIC are "Female educational level," "Female employment,"



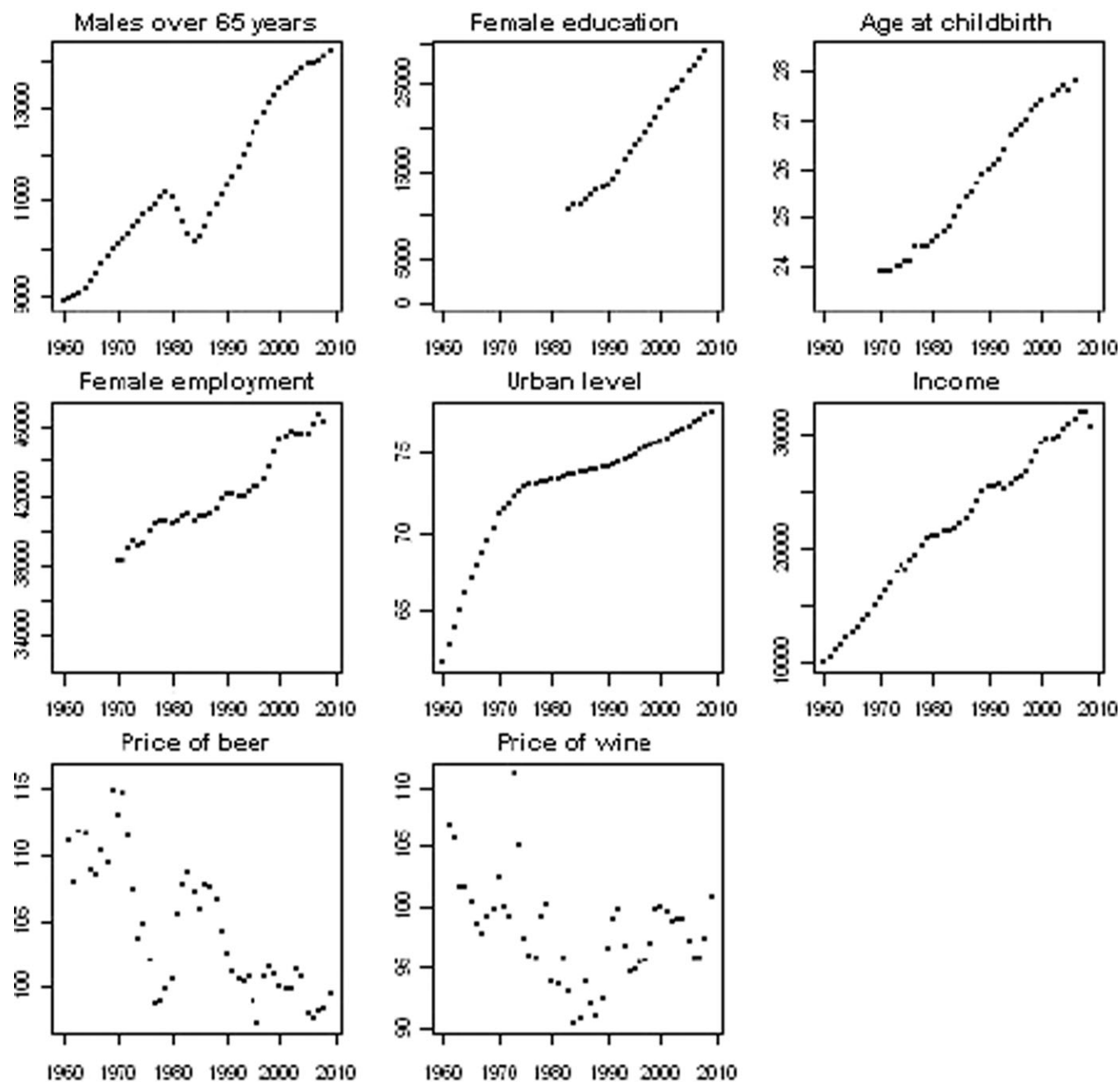


FIGURE 4. Unplanned variables selected for the analysis, France. Data from different national sources, France. Abscissas: Years; Ordinates: Male 65 years old and over: Rate per 100,000 population; Female education: Number of F who have completed tertiary education per 100,000 F population; Age at child birth: Mean age of women at their first childbirth; Female employment: number of F 15–64 employed; Females per 100,000 F population 15–64; Urban level: Number of residents in large town per 100,000 population; Income: Average income per household; Price of wine: Average price of wine; Price of beer: Average price of beer.

“Mothers’ age at all childbirths,” and “Urban level.” These variables were retained for next analysis.

## RESULTS, COMMENTS, AND DISCUSSION

### Association Between the Selected Unplanned Variables and Alcoholic Beverage Consumption

Table 2 shows the correlations of the French values for the four socioeconomic unplanned variables selected for the analysis. The reported correlation is the mean among the correlation estimated in each imputed dataset. Values

larger than .70 and smaller than  $-.70$  indicate a strong correlation.

Since these four variables turned out to be highly correlated, in general, a separate analysis for each variable was carried out. Therefore, four different models were estimated for each of the selected indicators. The lagged values of these variables (mean value of the unplanned indicator in the year under study up to two years before) were inserted in the analysis.

Since the analysis was carried out on the data resulting from a multiple imputation procedure, each model was



TABLE 2. Correlations among the unplanned variables selected for the analysis (France)

|                           | Female education | Female employment | Mother's age at childbirth | Urbanization | Income |
|---------------------------|------------------|-------------------|----------------------------|--------------|--------|
| Female education          | 1.00             | .93               | .86                        | .85          | .92    |
| Female employment         | .93              | 1.00              | .93                        | .93          | .99    |
| Mother age at childbirths | .86              | .93               | 1.00                       | .77          | .94    |
| Urbanization              | .85              | .93               | .77                        | 1.00         | .93    |

estimated on a different imputed dataset (five datasets). The results of the models summarized on the five imputed datasets are reported in Table 3.

A positive coefficient indicates that an increase/decrease in the explanatory variable is associated with an increase/decrease of the total alcoholic beverage consumption. A negative coefficient indicates that an increase/decrease in the explanatory variable is associated with a decrease/increase of the total alcoholic beverage consumption.

The Confidence Interval (CI) was set at .90, which is a liberal use of CI, in order to capture even minor information from data, instead that .95, which is the usual value for CI.

Then the AIC for each model was calculated. The AIC is a measure that accounts for goodness of fit and parsimony of the model. By and large, lower AIC values indicate better models.

As shown in Table 3, in France the model with "female employment" generates the minimum AIC, i.e. it is a better model for the interpretation of results. This suggests that the increase in female employment during the study period (see Figure 4) was associated with about a 2.5% increase in alcohol consumption. The model with "mother's mean age at all childbirths" is negatively correlated with alcoholic beverage consumption and is associated with a 3.5% decrease in alcohol consumption. The remaining two models are not statistically significant.

While GISAH data, which refer to alcoholic beverage total consumption, show that consumption in France has markedly decreased over the study period, this statistical analysis documents that female employment positively correlates with total consumption of the French general

population (not with women's consumption). Instead, if we look at Figure 1 (consumption data, which decreases) and Figure 4 (female employment, which increases over time), we would expect that in the statistical model the two phenomena would manifest opposite trends, i.e. be inversely correlated. This is not the case. Furthermore, Table 3 shows that the correlation coefficient between female employment and mother's age at childbirth is .93, highly positively correlated, as it can be seen with the two curves in Figure 4. The finding that they significantly contribute to alcoholic beverage consumption in opposite ways is very confusing. Maybe it is the high correlation between *all* of these variables (female employment, female education, mother's age at childbirth, income, urbanization, time. . .) that "produces" such a result. A caveat to consider that retaining the female employment variable for the model, which appears to be correlated to alcoholic beverage consumption in the "wrong" way may be misleading. The model chosen and the results it will produce have to be interpreted very cautiously if they are used for needed intervention planning and implementation.

However, the results that an increased age of mothers at childbirths correlates with a decrease total consumption are in keeping with the study's derived observation. Because of their new roles in modern society, women would wait more than in the past before becoming mothers, and overall this would be the signal of a society which has become industrial and urbanized as well as turning to services, maintaining less of the traditional behavior of drinking wine in greater amounts, which was specific of the rural culture, where also genders were kept more separated. See also Sulkunen and his sociocultural interpretation of the decrease of wine drinking in France (Sulkunen, 1989).

TABLE 3. Regression coefficients describing the relationship between 15+ per capita recorded alcoholic beverage consumption (Source: WHO 2009) and four selected unplanned indicators in France\*, with 90% CI, and AIC values\*\*.

| France                        | Coefficient | CI90           | AIC     |
|-------------------------------|-------------|----------------|---------|
| Female education              | .04         | (-.01, .1)     | -170.86 |
| Female employment             | 2.46        | (2.03, 2.9)    | -207.29 |
| Urbanization                  | .21         | (-.65, 1.07)   | -167.99 |
| Mother's age, all childbirths | -3.52       | (-4.75, -2.29) | -179.82 |

\*results from four separate regression models for each unplanned factor, adjusted for time trend, income, proportion of males >65 of age, and prices of beer & wine

\*\*lower AIC values indicate better models which explain consumption changes.

### Association Between Control Policy Measures and Alcoholic Beverage Consumption Adjusted for the Unplanned Variables

Table 4 presents the results of the models for single policies summarized on the five imputed datasets. Since the policies are analyzed one at time, the effect estimated for each policy estimates the changes in the mean total alcoholic beverage consumption values before and after the policy implementation, resulting in an underestimation of the policy effect.

A positive coefficient indicates that the occurrence of the policy measure is associated with an *increase* of the total alcohol consumption. A negative coefficient indicates

TABLE 4. Correlation coefficients of 15+ per capita recorded alcoholic beverage consumption (source: WHO, 2009) and selected policy measures in France\*, with 90% CI\*

| France   | Coefficient | IC 90          |
|----------|-------------|----------------|
| Policy 1 | .14         | (.09–.18)      |
| Policy 2 | .14         | (.09–.18)      |
| Policy 3 | –           | –              |
| Policy 4 | –.11        | (–.15 to –.06) |
| Policy 5 | –.18        | (–.21 to –.15) |
| Policy 6 | –.11        | (–.15 to –.06) |

\*effects adjusted for lower AIC unplanned indicator (female employment).

that the existence of the policy measure is associated with a *reduction* of the total alcohol consumption.

The coefficient is the logarithm of the differences between consumptions after and before each policy measure. Larger figures indicate larger effects. Each coefficient estimated for the policy effect should be interpreted together with its CI. The latter is a measure of the variability of the estimates and gives a set of possible values for the coefficient values. This means that if the CI contains the value 0 (i.e. it is between a lower negative value and an upper positive value), then the absence of any effect of the policy is plausible. If on the contrary the CI is placed above, or be-

low zero, the policy measure has, respectively, a positive or negative effect. (Figure 5).

The results of the analysis on control policy measures (see Table 4 and Figure 5) show that both 1970 measures, .8 g/L BAC policy and creation of “nutritional hygiene centers”, were associated with a 14% increase of alcoholic beverage consumption. On the contrary, control measures introduced in 1987, 1991, and 1995 (TV ads ban; forbidden alcohol sale to minors and Loi Evin’s ads control; .5% BAC measure) are associated with a decreasing trend in alcoholic beverage consumption over time, by about 11%, 18%, and 11% respectively.

Due to very strong lobbying of wine producers, the price of wine (which has always been the most common alcoholic drink in France) has always been low. The price of other drinks is higher in comparison with their alcohol content, but as they represent a small proportion of what French people drink, the impact of price is not very important. As a consequence, it would be reasonable to conclude that alcoholic beverage prices, in France, have really been used as a control policy.

The only exception concerns alcopops. A high tax was imposed on them just after they appeared on the market, in order to protect young people who were the main market target for these drinks.

While the negative coefficient for the restrictive policies 4–6 indicates that their presence is associated

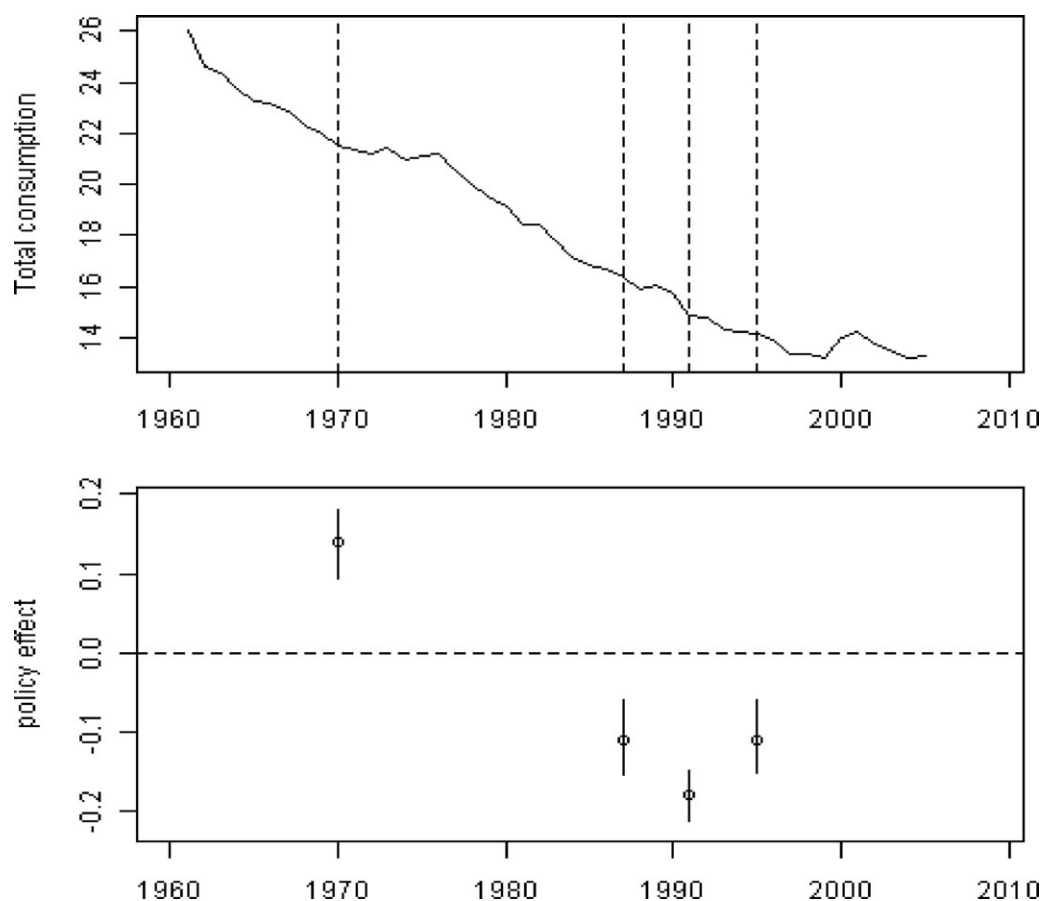


FIGURE 5. Upper graph: total alcohol consumption. The vertical dotted lines indicate the occurrence of policy measures. Lower graph: estimated policy effect with 90% CI (France).

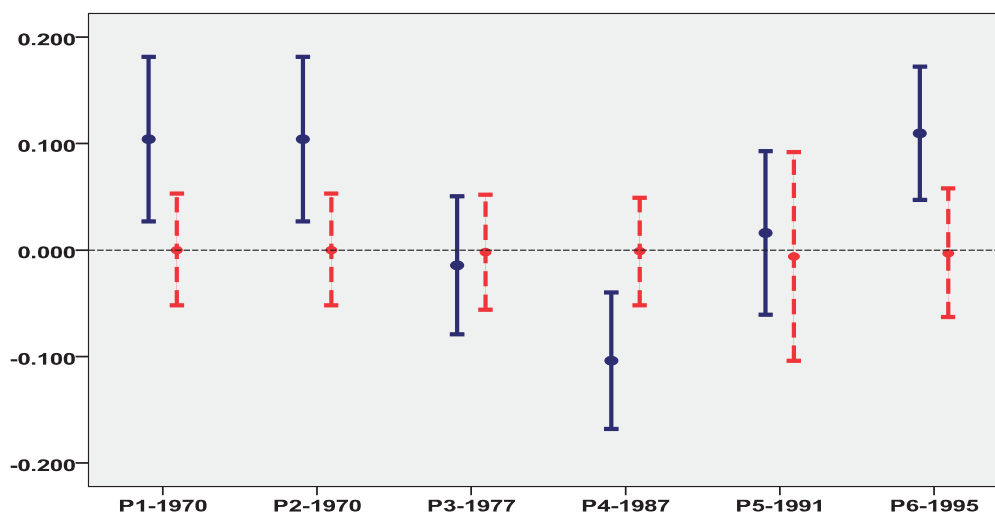


FIGURE 6. Correlation coefficients of direct and indirect liver deaths\* and selected policy measures and their year of introduction in France\*\*, with 90% CI. (continuous line = direct effect and dotted lines = indirect effects); \**direct* = policy effect on harm and *indirect* = policy effect on harm mediated through consumption effect; \*\*effects adjusted for lower AIC unplanned indicator (female employment).

with a reduction in alcoholic beverage consumption—suggesting that these policies might have contributed to the on-going alcohol drinking decrease in France—control policies 1 and 2 are unexpectedly positively correlated with consumption. In particular, one would expect that the effect of policy 1 should be similar to that of policy 6; a negative effect on consumption, both being drinking and driving control measures. In the graph, it seems that the curve decreases all the time, so the positive effect of both 1970 measures, since on alcoholic beverage consumption, is not obvious. One might consider that the stagnation observed from 1970 until about 1976 represents the brake in alcoholic beverage consumption that the 1970 measures could have effected. As noted earlier, the

study's results have to be considered and used with caution, especially because of correlation problems and the choice of the female employment variable as an adjusting variable.

#### Association Between the Selected Control Policy Measures and Harms Related to Alcohol Consumption

Figures 6 and 7 describe the correlation between direct and indirect control policy measures and liver mortality (1970–2008), and transport mortality (1979–2008), respectively.

Figure 7 shows that 1987 total TV ads ban is significantly associated with a direct decrease in liver mortality,

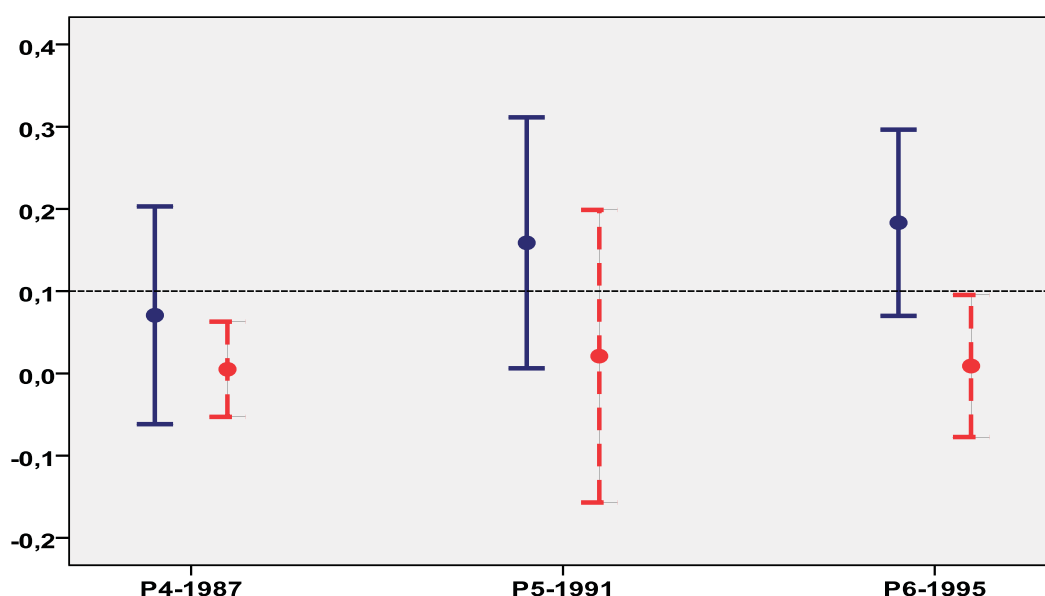


FIGURE 7. Regression coefficients of direct and indirect transport deaths\* and selected policy measures and their year of introduction in France 90% CI. (continuous line = direct effect and dotted lines = indirect effects); \**direct* = policy effect on harm and *indirect* = policy effect on harm mediated through consumption effect; \*\*effects adjusted for lower AIC unplanned indicator (urbanization).

but that 1970 and 1995 drinking and driving measures directly relate to an increase of liver mortality.

Figure 7 indicates that, while policy measures appear to have no significant direct effect on transport accident mortality, the 1987 total TV ads ban and the 1995 BAC law are significantly associated with a consumption-mediated drop in transport mortality.

The role that French policy measures have had on the reduction of alcohol mortality for liver disease and cirrhosis, and for transport deaths, during the study period, appears to be unclear. The 1987 TV total alcoholic beverage ads ban was associated with reducing liver mortality, as well as a consumption-mediated effect on reducing transport deaths, while the 1995 BAC measure demonstrated a consumption-mediated tendency toward the reduction of deaths due to transport accidents. However, all the other policies show no relevant impact, or even an unexpected death increase, like the 1970 measures, which appear to have a positive direct effect on liver deaths. It is not very relevant to talk about a direct effect of drink-driving measure on liver deaths. Perhaps the limited timeframe made it difficult to properly evaluate the harm done by alcohol consumption, especially on the liver. Again, these results have to be read, considered, and used with caution. Significance here (or what it suggests) has to be questioned and, in any case, coefficients are low (about .1).

Notwithstanding, the results related to the 1987 and 1995 policy measures, may be in keeping with the results of the first analysis which showed that the 1987–1995 policies were associated with decreasing alcoholic beverage consumption.

### Study Limitations

A number of limitations to the generalizability of the study's results merit consideration.

The study was not designed to explore the necessary conditions (micro to macro levels, individual and systemic-related, temporary or ongoing, among others), for the selected unplanned factors, *post facto*, and their relevant dimensions and qualities, to operate (to begin, to continue, to change as realities change, to cease, to begin again, etc.) or not to operate.

Control measures, (edicts, regulations, laws, traditions, norms, values, etc.) local or national, depend upon the “abidingness,” or lack of individuals and or systems (family, community, etc.) in a range of temporary or more permanent, stable, random, and chaotic roles, conditions, situations, and environments.

The planning, delineation, effective implementation, and assessment of any control policies, enabling as well as interfering with them, are associated with a range of individual and systemic stakeholders; each with their unique sources of influence, levels, and qualities of permanence or temporariness, agendas, goals, etc. Effecting alcoholic beverage consumption is no different; it too has its stakeholders. Some are known, some are as yet unknown and some may be unknowable; measureable as well as unmeasureable.

The AMPHORA study was not designed to measure these aspects of control factors, as well as others which you, the reader, may be aware of. These study limitations offer a rich opportunity for further needed study.

### Summary and Implications for Alcoholic Beverage Control Policies in France

The French socioeconomic and political scene during the period 1960–2000s is characterized by a range of “big events”<sup>3</sup> as well as less “big events.” These included the presidency of De Gaulle in the 1960s, the student revolution and social conflicts between the end of 1960 and 1970. During the 1980s, media were liberalized, and a wave of nationalizations (EXAMPLES) was approved. In 1995, liberal (in French, free market) reforms were introduced. Increased immigration (legal and illegal) with its range of unresolved challenges occurred. France has been part of European Union since its foundation as EEC in 1957.

During the 1950s, France manifested its post-WW II peak in alcoholic beverage consumption. Since then a decrease started, which is also observable at the start of our study period, when in 1961 total alcoholic beverage consumption was 26.03 L of pure alcohol per capita of people 15 years old or more. A constant decreasing trend has been documented (substantially due to decreased wine consumption) that reached 11.55 L in 2007, a drop of about 56%.

France is traditionally a “wine country,” wine consumption being 79% of the total alcoholic beverage consumption in 1961 and to 64% in 2007. During the same period, beer drinking also decreased by a slower steady pace of 26%, while spirits consumption manifested an up and down drinking trend, ending with a smaller drop of 9%. Spirits drinking placing second after wine.

The study's mortality indices, (1) the 1/4 decrease in liver disease-related deaths, and (2) the drop in transport related-accidents' mortality by about 1/3 during the period 1979–2008, are paralleled by the decrease in alcohol (wine) consumption, which has occurred in France between 1960 (and also during the previous decade), and during the 2000s.

The analysis, which has been done in this project, documents that the decrease of alcohol consumption (more specifically wine consumption) is related to both selected social and economic events occurring between 1960 and the 2000s, like increased female employment, income, and urbanization. All these factors are indicators of the transformative aspects of societal, cultural, political, religious-spiritual, media-associated and economic

<sup>3</sup>This relatively new term, introduced into the intervention literature, by Friedman et al. (Samuel R. Friedman, Diana Rossi, Peter L. Flom. (2006). “Big events” and networks: Thoughts on what could be going on. *Connections* 27(1): 9–14.) refers to major events such as megadisasters, natural, as well as man-made, famine, conflict, genocide, disparities in health, epidemics, mass migrations, economic recessions, etc. which effect adaptation, functioning and quality-of-life of individuals as well as systems. Existential threat, instability, and chaos are major dimensions and loss of control over one's life is experienced. Editor's note.

change on individual, family, social network, community, and national lifestyles and qualities-of-life, one of which is wine drinking in France, as was pointed out by previous research.

Wine drinking also appears to be associated with restrictive measures when they were introduced, which constituted an addition to the effect of the study's selected contextual factors, as it has been reported above.

The policy measures studied do not appear to have any strong effect on alcoholic beverage consumption-related harms, for which, it can be reasonably assumed, that they were originally legislated.

### Declaration of Interest

The authors declare no conflicts of interest. The authors alone are responsible for the content and writing of the article.

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